

Lightweight Low-Light Image Enhancement Techniques for Edge Devices

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Abstract—The field of computer vision has a lot of practical applications in today's world, such as surveillance, autonomous navigation, medical imaging, etc. However, these computer vision systems frequently encounter issues such as images captured under poor lighting conditions, low contrast, backlit images, and noise infiltration into the image. The enhancement of these dark and dimly lit images is crucial for maintaining the reliability and accuracy of computer vision systems. This paper provides a comprehensive overview of traditional and deep learning-based methods for low-light image enhancement. We thoroughly explore traditional methods such as grey level transformation, gamma correction, histogram equalization, and Retinex theory-based methods, highlighting their effectiveness and limitations. In addition, the paper discusses the suitability of the methods for edge deployment. The paper also proposes a model for the detection and classification of low-light images on edge devices.

I. INTRODUCTION

Computer vision has many applications, from autonomous navigation and surveillance systems to medical imaging. One of the major challenges the domain faces right now is having images with low visibility and brightness. Images and videos taken under poor lighting conditions, backlit images, and uneven lighting cause several problems like poor visibility, reduced accuracy in object detection, unreliable feature extraction, etc.

Our work provides an overview of the various methods for enhancing dark and dimly lit images and frames. Our work explores the various traditional low light image enhancement techniques such as Gray transformation methods, Histogram equalization methods, Gamma correction techniques, and Retinex-based methods [19].

The paper also explores various deep learning-based image enhancement methods [10], classified into supervised, unsupervised, and zero-shot learning techniques, depending on the learning algorithm used. The first part of the paper introduces traditional image enhancement methods like gray-level transformation, Gamma correction, histogram equalization, and Retinex-based methods. Under deep learning-based methods, we first discuss the various supervised learning enhancement algorithms such as LLNet [11], Retinex-Net [22], KinD [25],

and MBLLEN [13], and you only need one color space [2]; then, we discuss the EnlightenGAN [6] method, which falls under the unsupervised learning technique, followed by zero shot methods, namely, Zero-DCE [3] and RRDNet [26].

II. LOW LIGHT IMAGE ENHANCEMENT METHODS

Image enhancement techniques for poorly illuminated images can be categorized into several main groups: conventional approaches such as Gray Transformation, Histogram Equalization, and retinex methods, as well as Deep Learning techniques [7]. These categories can be further subdivided based on their operations' distinct principles.

A. Gray Level Transformation Methods

Gray Level Transformation methods are image transformation techniques [1] that are used to enhance dimly lit images by changing the gray value brightness levels of individual pixels. These methods apply a mathematical function to map each pixel's grayscale intensity to a new value, effectively adjusting brightness and contrast to improve visibility. These methods apply global transformations, making it difficult to handle uneven lighting conditions and may also result in limited enhancement of severely dark regions. These methods are highly suitable for Image enhancement in edge devices, due to simple memory and computational requirements.

1) *Linear transformation*: Linear gray-level transformation techniques are fundamental image processing operations in which a linear mathematical function maps the values of the input pixel to the required pixel intensities [20]. Linear transformation methods use mathematical functions to adjust pixel intensities and enhance low light images. These methods modify the gray levels of an image to improve brightness and contrast, providing a simple and computationally efficient approach to low-light image enhancement.

Some examples of linear transformation methods used are negative transform, contrast stretching, etc.

Since the Linear Gray level transformation techniques such as contrast stretching and negative transform only uses only pixel-wise operations, it has high computational efficiency and

minimal memory usage, making it highly suitable for edge devices. However, Linear methods often fail to preserve local contrast or adapt to non-uniform lighting.

2) *Nonlinear transformation*: The main concept of nonlinear gray transformation is that it modifies the gray values of an input image using a non-linear function. Commonly applied functions for this transformation include logarithmic, gamma correction, and other enhanced variations.

Nonlinear gray-level transformation techniques, such as logarithmic and gamma corrections, use nonlinear curves, which allows localized enhancement, especially in dark regions. However, they can amplify noise in low-intensity areas and typically require post-processing normalization. Despite these shortcomings, their flexibility and improved output quality make them practical for lightweight enhancement on resource-constrained edge devices.

Logarithmic Transformation

A logarithmic transformation, which falls under Gray Transformation methods, employs a logarithmic function to map pixel values from the input to the output image.

This method obtains the output image at a particular pixel position by applying a logarithmic transformation to the input image. The transformation also involves scaling the input by a control parameter c [22]. The parameter c regulates the enhancement intensity, allowing fine-tuning of contrast and brightness in the resulting image. This method is suitable for real-time use; in addition it enhances low intensity pixels. It can amplify noise in dark regions, and requires normalization and scaling after transformation.

Gamma Correction Techniques

The Gamma correction transformation technique [20] enhances low-light images by applying a gamma transformation function. This function utilizes a gamma parameter to adjust pixel intensities, effectively improving visibility in dark regions while preserving contrast in brighter areas.

This method offers good performance for low and dimly lit image enhancement through its computationally efficient operation that requires minimal resources, making it particularly suitable for edge devices and resource-constrained devices. The adjustable gamma parameter provides flexible enhancement capabilities, allowing brightness and contrast optimization based on specific image characteristics, while maintaining structural integrity by preserving image details more effectively than linear methods without introducing excessive distortion in moderately illuminated regions.

Despite its computational advantages, the gamma correction exhibits several limitations that constrain its effectiveness in challenging low-light scenarios. This technique requires careful tuning of the gamma value, which can affect the detail and contrast preservation. It can exacerbate noise in dark regions, particularly in low-quality images, degrading overall quality. So, even though Gamma correction is well-suited for applications requiring fast and simple low-light image enhancement, it is less effective in highly variable or

extreme low-light conditions compared to advanced Retinex or deep learning approaches.

B. Histogram Equalization Methods

Histogram Equalization methods [20] are utilized to enhance image contrast. An image histogram visually depicts the distribution of pixel intensity values. This enhancement approach redistributes frequently occurring intensity values, improving global contrast and enhanced images.

Histogram equalization methods can be classified into global histogram equalization(GHE) and local histogram equalization(LHE) based on the area and range of the image in which the algorithms are applied [21].

Global histogram equalization

Global histogram equalization(GHE) is the easiest and most commonly used histogram equalization algorithm. GHE technique improves the overall contrast of an image by redistributing its intensity values across the entire range of the image. GHE operates on the entire image, transforming the histogram of pixel intensities into a more uniform distribution, often resulting in improved visibility of details, especially in dark or low-contrast areas. Although this technique has its advantages, such as computational efficiency and ease of implementation, it also faces disadvantages like over-saturation and loss of local details in the image.

Because of the simple, non-iterative and memory light design and efficient implementation of this technique, it is well-suited for edge devices with limited computational power and memory. Since GHE is deterministic and requires no learning or prior data, it is suitable for standalone edge deployments too. However, its global nature can lead to over-saturation and loss of local details.

Local histogram equalization

The Local histogram equalization technique [14] divides the image into small regions, computes a local histogram for the region, and then applies the required transformation function to each image block. This allows the contrast of each block to be optimized independently, which helps reduce the over-saturation and retains the image details. Although this method offers better visual output than global methods, its higher computational and memory demands may limit its suitability on resource-constrained edge devices, especially for real-time or battery-powered applications.

Adaptive histogram equalization (AHE)

The AHE technique segments the image into blocks and applies distinct transformations to each region based on local contrast. This approach independently enhances contrast in various image areas, simultaneously improving dark and bright regions.

This method provides a strong local contrast enhancement by equalizing pixel intensities within the divided image blocks. Since it is a deterministic, non-learning-based method that doesn't require external training or models, it is great for offline and embedded systems. But, AHE's block wise histogram

calculation and pixel remapping are compute heavy, especially for high resolution images or real time use, making it not viable for real time or low power edge devices.

Contrast limited adaptive histogram equalization (CLAHE)

The CLAHE technique [5] enhances contrast by dividing the image into smaller blocks and applying distinct transformations to each block based on its local contrast. This approach helps reduce artifacts and improve the overall image quality. Additionally, CLAHE limits the amount of contrast applied to each block by setting a user-defined threshold for the contrast value. Adjusting this threshold allows the user to control the degree of contrast enhancement applied to each region. This method effectively reduces the block effect typically introduced by dividing the image into smaller regions, as seen in Adaptive Histogram Equalization (AHE).

This technique enhances local contrast with reduced noise and improved visual continuity by limiting and interpolating contrast per region. Its tunable parameters and efficient implementations make it a practical solution for edge devices, balancing enhancement quality with performance, especially in real-time low-light applications.

C. Retinex Methods

Retinex-based enhancement techniques [20] aims to improve the visibility of poorly illuminated images and video frames by leveraging the principles outlined in Retinex theory [24]. According to this theory, an image can be interpreted as the combination of its illumination and reflectance components, where the observed color or intensity is modeled as the product of these two factors. According to Retinex theory [24], if the illumination component of the image can be accurately estimated from the image intensity, then the reflection component can also be isolated [20]. This decomposition allows for the suppression of illumination-related distortions in the image, thus contributing to improved visual quality. By emphasizing on the reflectance characteristics and diminishing the impact of inconsistent lighting, the method significantly boosts visibility in dimly lit environments.

Single-scale retinex method

The Single-Scale Retinex(SSR) model operates [20] by estimating the illumination component of an image using a single Gaussian filter and then removing it to obtain the reflectance component, which represents the enhanced image. The illumination component here refers to the lighting conditions captured in the image, separate from the spatial features of the image. The algorithm works by estimating the illumination component.

Since the SSR uses a single Gaussian filter for estimating illumination, it becomes less complex and computationally efficient. Being a classical image processing method, SSR doesn't require external data, training, or complex memory resources, making it a good fit for resource constrained devices. However, it provides limited enhancement capacity and is sensitive to parameter selection, making it better suited for moderately powerful edge devices.

Multi-scale retinex method

The Multi-scale retinex (MSR) technique operates by decomposing an image [15] into multiple scales and applying a center/surround retinex function to each. The scale here is the area considered around each pixel when applying the retinex function. The center refers to a pixel or small region in the image and surround, the neighboring area around that center. This process effectively separates the illumination from the reflectance components of the image.

This multi-scale approach offers significant advantages over single-scale methods. By integrating information from different spatial extents, MSR achieves comprehensive image enhancement. It effectively amplifies fine details and boosts overall contrast while maintaining color consistency across the image.

MSR offers superior enhancement by using both local and global image statistics on multiple spatial scales. It is highly effective for low-light image restoration, but its computational intensity makes it impractical for most real-time edge applications unless carefully optimized or hardware accelerated.

D. Deep learning based methods

1) Supervised learning methods:

LLNet

The LLNet [11], a deep autoencoder-based approach, enhances low-light images by enhancing contrast and reducing noise in the images that are synthetically darkened. This model leverages a stacked sparse denoising autoencoder to enhance the input images by improving the brightness and suppressing noise. Using a series of deep neural networks, it is able to improve natural images captured in low-light conditions. LLNet [11] operates on principles similar to Sparse and Low-rank Denoising Autoencoders (SSDA), leveraging their denoising capabilities and the ability to learn complex patterns in low-light image data.

The framework makes use of synthetically generated dark images, which reduces the need for paired training data. Despite the aforementioned advantages, the method struggles to attain optimal brightness levels in extremely low light scenarios, and the architecture needs significant memory and computational needs, making it unsuitable for edge devices and in computationally-constrained devices. The heavy computational requirements also make it infeasible for real-time processing.

Retinex-Net

The Retinex-Net framework [22] enhances low-light images based on the Retinex theory, which separates the input image into reflectance and illumination components. This decomposition into the components preserve the actual colors and textures of the objects in the image while improving the lighting this makes the resultant images look more natural and realistic. Since the model is extensively trained with paired training data, the model enhances low-light images very effectively. The Retinex-Net provides consistent and accurate

results on datasets such as LOL dataset [23] and test set introduced by the LIME method [4].

Despite the consistent enhancement results of this model, the dependency on the paired data for training can pose a challenge in specific situations where collecting such data is difficult. The method can amplify noise and other artifacts in noisy low-light images. The dual network architecture for the decomposition and the enhancement network demands high memory and computational abilities, limiting its suitability for edge devices compared to methods like Zero-DCE [3].

Kindling the Darkness (KinD)

The KinD network is a low light image enhancement method [25], which works based on the Retinex theory principle [17]. The designed method also has a mapping function for dynamically modifying the brightness levels of the image according to the user's preference. The network also contains a module to remove undesirable features in the image, such as noise or other disturbances that impact the image quality.

Since this technique ensures robust decomposition of illumination and reflectance components, the model is more effective in enhancing low-light images with high noise and near-zero illumination. The model, enables users to control brightness levels, giving more leverage on the resulting enhanced images. Since the model is trained with paired images under different light settings, it reduces the dependency on ground-truth data, which may be difficult to obtain in certain situations. Despite all the aforementioned benefits of the model, the retinex-based decomposition causes darker output images compared to methods like Zero-DCE [3], MBLLEN [13], You Only Look at One Color Space [2] and EnlightenGAN [6]. The framework architecture has separate networks for decomposition, reflectance and illumination increases computational demands of the system, making it less efficient than models like Zero-DCE [3]. Due to these limitations, this is not a recommended framework for Low-Light image enhancement on edge devices. However, this can be made feasible for edge devices by integrating it with dedicated specialized hardware accelerators.

Multi-branch low-light enhancement network (MBLLEN)

The MBLLEN [13] is a fully convolutional neural network designed to improve dark and poorly lit images and videos. It consists of three types of modules: the feature extraction module (FEM), the enhancement module (EM), and the fusion module (FM) [12]. FEM extracts features at various levels, while EM enhances these extracted features. The final output is produced through multi-branch fusion by the FM.

The robust feature extraction module enables the model to extract both local and global image spatial features, making the resultant image more accurate and natural. The enhancement is robust and consistent because of the multiscale architecture that captures spatial features effectively and combines multi-scale information through feature fusion. The model also has End-to-End Learning, which leverages paired datasets to ensure optimized enhancement and spatial consistency.

This framework lacks separate denoising mechanisms, which can cause noise amplification and artifacts in noisy inputs. Because of the End-to-End learning, there is a dependency on obtaining a high-quality paired dataset, which can be hard to obtain, especially in low-light scenarios. The multi-branch architecture also significantly increases memory and computational demands, compared to lightweight models.

Due to the high computational complexity and the memory consumption, the technique is impractical for edge devices and other hardware devices that do not have dedicated accelerators. Therefore, lightweight alternatives such as Zero-DCE [3] and LLNet [11] are more suitable for deployment on edge devices.

You Only Need One Color Space

Most existing LLIE methods use the sRGB space to define relationships that map the low-light to enhanced image mapping. However, there is a strong interdependence between color and the image brightness in the sRGB color space. Thus, a slight disturbance in the color will cause noise in the illumination and color of the image. To address this issue, the paper [2] introduces a new learned color space, the Horizontal/Vertical Intensity(HVI) space. The HVI space separates the luminance from the color information.

The framework, also uses a method, known as the Color and Intensity Decoupling Network (CIDNet) [2]. The CIDNet has two branches, namely, the HV branch, and the intensity branch. The HV-branch in the CIDNet is used to suppress the noise and handle the chromaticity in the dark regions and I-branch, to estimate the brightness in the whole image. Here, the computational complexity is reduced significantly compared to other methods that leverage multiple color spaces. Since the architecture combines both convolutional and transformer layers to learn the spatial data, the enhancement is more accurate and natural.

However, the method does not perform well with extremely dark or poorly illuminated inputs. In contrast, multi-stage decomposition techniques such as RetinexNet [22], KinD [25], MBLLEN [13] demonstrate better performance. This method also heavily depends on paired low-light and well-lit image datasets, which can be difficult to obtain. The framework's transformer-based architecture [18] increases computational complexity, making it less suitable for edge devices and resource-constrained devices compared to lightweight models like the Zero-DCE.

2) Unsupervised methods:

EnlightenGAN

EnlightenGAN uses a generative adversarial network(GAN) to enhance [6] dimly-lit images. EnlightenGAN eliminates the need for having corresponding normal-light image pairs by using an unsupervised GAN. It uses a global-local discriminator structure that helps distinguish between enhanced and real images. The method uses an attention-directed U-Net generator [16] that enhances the image. The attention mechanism present in the architecture enables the network to focus on low-light regions specifically, enabling the model to

enhance dark regions selectively, while preserving the well-lit regions.

As the network does not need paired training data, it becomes suitable in situations, where collection of such data is difficult. The self-regularized attention mechanism targets regions that are considerably dark, thus preventing over-exposure in already bright areas. The attention mechanism also aids in suppressing noise, thus creating cleaner output images.

3) Zero-shot learning methods:

Zero-DCE

Zero Reference Deep Curve Estimation(Zero-DCE) [3] introduces a zero-shot learning approach for low-light image enhancement, making it an innovative approach to enhance poorly lit images and video frames without relying on reference data. This model eliminates the need for paired training data or reference images, instead learning to improve low-light images by calculating pixel-specific tone mapping curves. This makes it highly practical for real-world applications where such data is unavailable. Zero-DCE employs a compact deep neural network to determine the optimal curve parameter maps for image enhancement based solely on the input image.

To enable zero-reference learning, the model utilizes various non-reference loss functions, including Illumination Smoothness Loss, Color Constancy Loss, Exposure Control Loss, and Spatial Consistency Loss, which collectively form the total loss [8].

The model's architecture is based on simple convolutional neural network (CNN) architecture, which avoids complex components like GANs or attention mechanisms used in methods like EnlightenGAN [6]. The model uses a series of shallow convolutional layers to predict enhancement curves, resulting in a small model size and low computational complexity. This reduces memory and power consumption, making this technique highly suitable for edge devices. Additionally, its zero-shot learning eliminates the need for on-device training or large pre-trained models, further lowering resource demands. Despite its advantages for edge devices, the method struggles with extremely dark images with noise. The Curve based enhancement is not as good at detail preservation compared to GAN-based approaches in high-contrast scenes.

RRDNet

Robust Retinex Decomposition Network(RRDNet) [26] is a zero-shot LLIE method. This approach utilizes a three-branch convolutional neural network to process images without requiring well-lit counterparts for training. The model operates by decomposing the input image into three distinct components: illumination, reflectance, and noise, following the principles of the Retinex model [9]. In RRDNet, well-lit data counterparts are not necessary to train the network. The training process involves iterative weight updates by minimizing a specially designed loss function.

Since the technique is a zero shot learning approach, it requires no paired training data, making it efficient for cases with no paired training data. The model also handles the noise

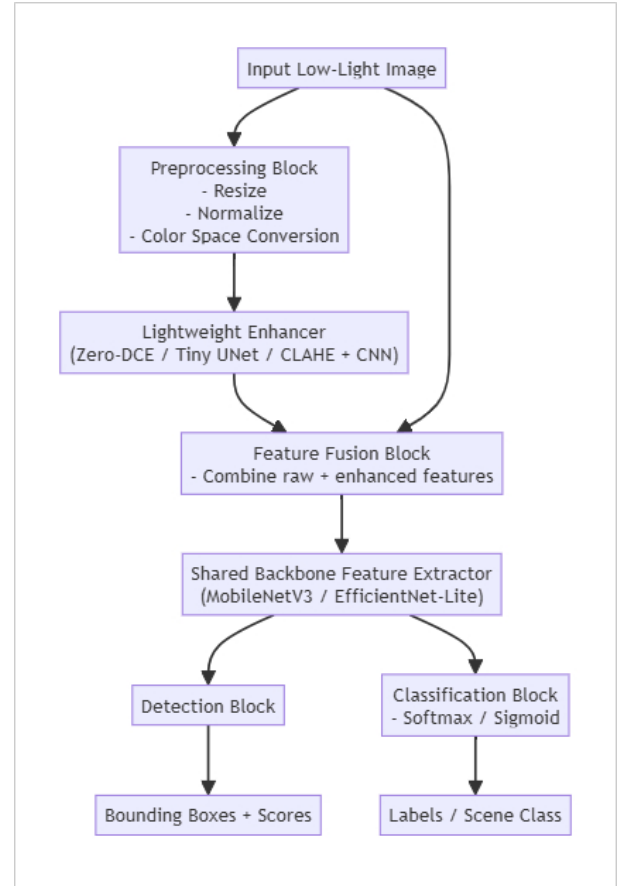


Fig. 1. Detection/Classification architecture for edge devices

in the image as a separate component, thus improving the noise handling better than similar methods like RetinexNet [22], EnlightenGAN [6], and MBLLEN [13]. While this approach offers better noise handling, enhancing with this model produces darker outputs compared to other methods mentioned here, limiting effectiveness in extremely low-light scenarios. Due to the decomposition process, the model can introduce a loss of sharpness, affecting the visual quality. The iterative decomposition process can increase the computational complexity, making it less suitable for edge devices compared to lightweight models such as Zero-DCE [3].

RRDNet [26] is more a denoising algorithm than a direct enhancement technique because it focuses mainly on denoising through residual learning and region-based processing, as opposed to the unsupervised learning techniques of EnlightenGAN [6] and Zero-DCE [3].

III. PROPOSED METHODOLOGY

The figure 1 shows the architecture that can be used to perform detection/classification tasks on low-light image inputs. Initially, the input is passed to a preprocessing stage that performs normalization, resizing and color space conversion. Then, it is passed to the lightweight enhancement module. Zero-DCE is a suitable candidate for the enhancement. Tiny U-Net also can perform well in edge devices. The CLAHE along

with a shallow CNN can also be used as an alternative. Instead of supplying the enhanced image alone, we can supply the features from the raw image to the feature extraction blocks. The feature maps from this layer can be fed into light weight models such as Yolo-Nano or SSD variants for detection. In case of classification, feature maps can be fed to classification networks.

IV. RESULTS DISCUSSION

TABLE I
BENCHMARK RESULTS FOR YOLOV8 MODELS WITH IMAGE
ENHANCEMENT

Model + Enhancement	Total Inference (s)	FPS	mAP50-95
YOLOv8n	1.09	117.08	0.4291
YOLOv8n + CLAHE	0.59	215.26	0.4108
YOLOv8n + Zero-DCE	0.65	196.18	0.4218
YOLOv8s	1.08	118.30	0.5713
YOLOv8s + CLAHE	1.20	106.60	0.5671
YOLOv8s + Zero-DCE	1.13	113.67	0.5406

The table I shows the performance of YOLOv8s and YOLOv8n with image enhancement methods. In the model, n stands for nano and s stand for small. Total Inference is the time taken (in seconds) to infer the entire dataset. FPS is the Frames Per Second achieved during inference. mAP50-95 is the Mean Average Precision calculated across IoU thresholds from 0.50 to 0.95 (standard COCO metric).

The benchmark graph shown in figure 2 illustrates the trade-off between inference speed (FPS) and detection accuracy (mAP50-95) across different YOLOv8 model variants (n and s) and enhancement methods (CLAHE, Zero-DCE). YOLOv8n shows higher FPS (up to 215 FPS with CLAHE) due to its lightweight architecture, making it suitable for real-time or embedded applications where speed is critical. YOLOv8s, being a larger model, provides better detection accuracy (mAP50-95 0.57) but at the cost of reduced FPS compared to YOLOv8n. CLAHE preprocessing consistently achieves the highest FPS for YOLOv8n (215.26) but shows slightly lower mAP compared to Zero-DCE in the same model. Zero-DCE enhancement offers a balanced trade-off, improving image contrast adaptively, leading to marginally better mAP than CLAHE for YOLOv8n.

V. CONCLUSION

This paper presents a comprehensive review of widely adopted low-light image enhancement techniques. The literature survey explores and analyzes several classical and modern methods for enhancing images captured under poor illumination conditions.

The study first examines traditional approaches, including gray-level transformation methods, gamma correction techniques, histogram equalization strategies, and Retinex-based algorithms. Variants of each technique are reviewed, with particular attention given to their underlying mechanisms and enhancement principles. Furthermore, the limitations of these conventional methods, such as detail loss, color distortion, and high computational complexity, are critically discussed.

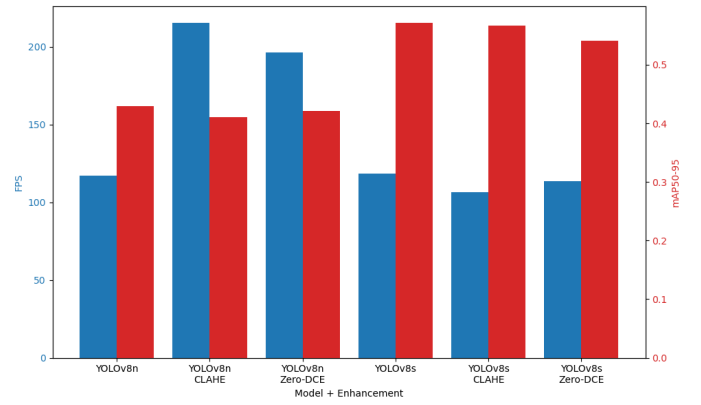


Fig. 2. Benchmark Results

In addition, the paper highlights the growing significance of deep learning-based approaches, categorizing them into supervised, unsupervised, and zero-shot learning paradigms. A review of their architectural frameworks and implementation strategies demonstrates their superiority in handling the complexities of parameter tuning inherent in traditional methods. Nonetheless, these approaches still face several challenges, including limited generalization across diverse lighting conditions, susceptibility to noise, and difficulty in accurately enhancing semantically complex regions. The paper also proposes a model for classification/detection in the edge environment. The comparison of YOLOv8 model variants (n and s) and enhancement methods (CLAHE, Zero-DCE) was compared.

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